

# Grade 11 – Term 1

## C5 Practice Activity 5 with Solutions

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### Evaluated criterion

**C5.** Calculates and interprets the mean and variance of continuous distributions using standard formulas and functions.

## 1 Introduction

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### Problems:

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- 1.1. Purpose and learning goals
  - 1.2. Continuous probability distributions
  - 1.3. Mean, variance, and standard deviation
  - 1.4. Formula reference
- 

Criterion C4 focuses on calculating and interpreting measures that describe the center and spread of a continuous probability distribution. The main measures used are the mean, or expected value, the variance, and the standard deviation.

These measures summarize different features of a probability distribution. The mean identifies its center, while the variance and standard deviation describe how widely values are distributed around that center. The appropriate formulas depend on the probability distribution and its parameters.

### 1.1 Purpose and learning goals

The purpose of this activity is to calculate and interpret the mean, variance, and standard deviation of continuous probability distributions using standard formulas.

For each situation:

1. Identify the probability distribution and its parameters.
2. Calculate the mean, or expected value,  $E(X)$ .
3. Calculate the variance,  $Var(X)$ .
4. Calculate the standard deviation,  $\sigma_X = \sqrt{Var(X)}$ .
5. Interpret the results in the context of the variable.



## 1.2 Continuous probability distributions

The distributions used in this criterion describe continuous variables. Their probability density functions assign probability to intervals through area under the curve. Different distributions have different shapes and parameters, so their means and variances are calculated using different formulas.

The distributions used in this activity are:

- Uniform
- Triangular
- Linear increasing
- Linear decreasing
- Trapezoidal
- Exponential
- Normal

The notation identifies the distribution and its parameters. For example,

$$X \sim U(a, b)$$

indicates a uniform distribution between  $a$  and  $b$ , while

$$X \sim N(\mu, \sigma^2)$$

indicates a normal distribution with mean  $\mu$  and variance  $\sigma^2$ .



## 1.3 Mean, variance, and standard deviation

The mean, written as  $E(X)$ , describes the center of a probability distribution. For a continuous random variable, it represents the long-run average value expected from repeated observations under the model.

Variance measures the spread of the distribution around its mean:

$$Var(X) = E[(X - E(X))^2].$$

The variance is expressed in squared units. Its square root is the standard deviation:

$$\sigma_X = \sqrt{Var(X)}.$$

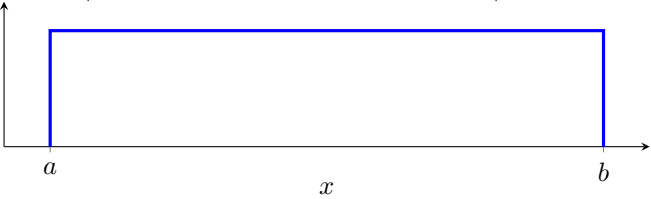
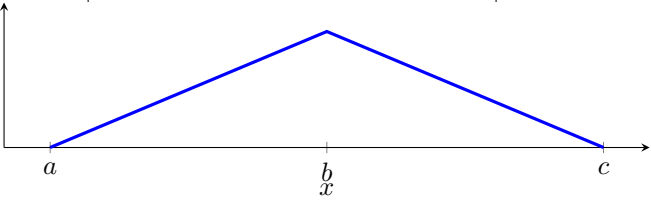
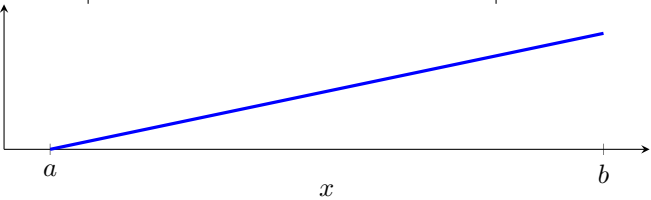
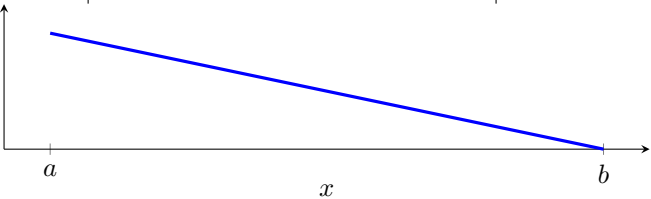
Unlike variance, the standard deviation has the same units as the original variable, which makes it easier to interpret in context.

For every problem, first identify the distribution and its parameters. Then select the corresponding formulas, substitute the parameters, calculate the results, and interpret their meaning using the units of the variable.

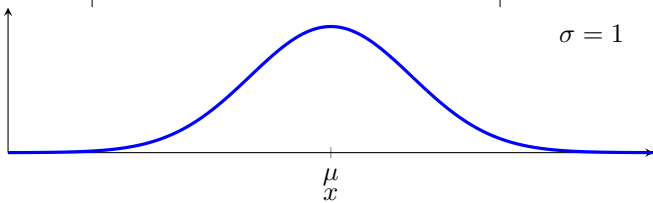
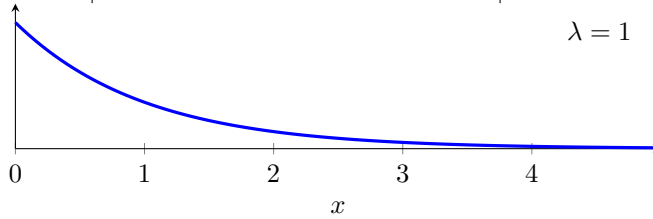
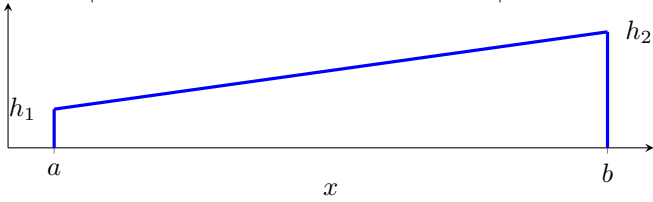


1.4 Formula reference

The following table summarizes the formulas used in this criterion.

| Distribution      | Notation                                                                                                                       | $E(X)$                | $Var(X)$                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------|
| Uniform           | $X \sim U(a, b)$<br>                         | $\frac{a + b}{2}$     | $\frac{(b - a)^2}{12}$                      |
| Triangular        | $X \sim \text{Tri}(a, b, c)$<br>             | $\frac{a + b + c}{3}$ | $\frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$ |
| Linear increasing | $X \sim \text{Lin}_{\text{inc}}(a, b)$<br> | $\frac{a + 2b}{3}$    | $\frac{(b - a)^2}{18}$                      |
| Linear decreasing | $X \sim \text{Lin}_{\text{dec}}(a, b)$<br> | $\frac{2a + b}{3}$    | $\frac{(b - a)^2}{18}$                      |

| Distribution | Notation                             | $E(X)$                                                                    | $Var(X)$                                                            |
|--------------|--------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| Trapezoidal  | $X \sim \text{Trap}(a, b, h_1, h_2)$ | $\frac{b-a}{3} \frac{h_1^2 + h_1 h_2 + h_2^2}{h_1 + h_2} + \frac{a+b}{2}$ | $\frac{(b-a)^2}{18} \frac{h_1^2 + 2h_1 h_2 + h_2^2}{(h_1 + h_2)^2}$ |
| Exponential  | $X \sim \text{Exp}(\lambda)$         | $\frac{1}{\lambda}$                                                       | $\frac{1}{\lambda^2}$                                               |
| Normal       | $X \sim N(\mu, \sigma^2)$            | $\mu$                                                                     | $\sigma^2$                                                          |



The standard deviation is obtained after calculating the variance:

$$\sigma_X = \sqrt{Var(X)}.$$

When interpreting a result, report the mean and standard deviation using the units of the original variable, while variance uses squared units.

The formulas above should be applied directly once the distribution and its parameters have been identified. The second table provides a compact reference without the graphs.

| Distribution      | Notation                               | $E(X)$            | $Var(X)$                                    |
|-------------------|----------------------------------------|-------------------|---------------------------------------------|
| Uniform           | $X \sim U(a, b)$                       | $\frac{a+b}{2}$   | $\frac{(b-a)^2}{12}$                        |
| Triangular        | $X \sim \text{Tri}(a, b, c)$           | $\frac{a+b+c}{3}$ | $\frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$ |
| Linear increasing | $X \sim \text{Lin}_{\text{inc}}(a, b)$ | $\frac{a+2b}{3}$  | $\frac{(b-a)^2}{18}$                        |

| Distribution      | Notation                               | $E(X)$                                                                        | $Var(X)$                                                              |
|-------------------|----------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Linear decreasing | $X \sim \text{Lin}_{\text{dec}}(a, b)$ | $\frac{2a + b}{3}$                                                            | $\frac{(b - a)^2}{18}$                                                |
| Trapezoidal       | $X \sim \text{Trap}(a, b, h_1, h_2)$   | $\frac{b - a}{3} \frac{h_1^2 + h_1 h_2 + h_2^2}{h_1 + h_2} + \frac{a + b}{2}$ | $\frac{(b - a)^2}{18} \frac{h_1^2 + 2h_1 h_2 + h_2^2}{(h_1 + h_2)^2}$ |
| Exponential       | $X \sim \text{Exp}(\lambda)$           | $\frac{1}{\lambda}$                                                           | $\frac{1}{\lambda^2}$                                                 |
| Normal            | $X \sim N(\mu, \sigma^2)$              | $\mu$                                                                         | $\sigma^2$                                                            |



## 2 Introductory Problems

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### Problems:

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- 2.1. Uniform distribution
  - 2.2. Linear increasing distribution
  - 2.3. Linear decreasing distribution
  - 2.4. Triangular distribution
  - 2.5. Exponential distribution
  - 2.6. Normal distribution
- 

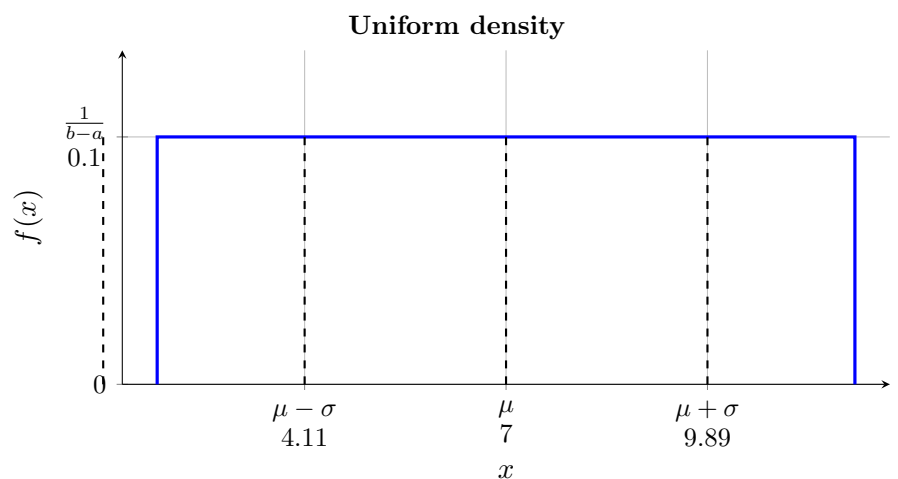
### 2.1 Uniform distribution

#### Distribution and parameters

A small business invests in a short-term opportunity. Based on similar investments, the money earned is expected to be between \$2,000 and \$12,000, with all values in this range considered equally likely.

Let  $X$  represent the money earned, measured in thousands of dollars. Therefore,

$$X \sim U(a, b) \quad \longrightarrow \quad X \sim U(2, 12) \quad \longrightarrow \quad a = 2, \quad b = 12.$$



#### Mean

For a uniform distribution,

$$E(X) = \frac{a+b}{2} \quad \longrightarrow \quad E(X) = \frac{2+12}{2} = 7 \quad \longrightarrow \quad E(X) = 7 \text{ thousand dollars}$$

## Variance

For a uniform distribution,

$$Var(X) = \frac{(b-a)^2}{12} \quad \longrightarrow \quad Var(X) = \frac{(12-2)^2}{12} = \frac{100}{12} = \frac{25}{3} \approx 8.33 \quad \longrightarrow \quad Var(X) \approx 8.33 \text{ thousand dollars}^2$$

## Standard Deviation

The standard deviation is

$$\sigma_X = \sqrt{\frac{25}{3}} \approx 2.89 \quad \longrightarrow \quad \sigma_X \approx 2.89 \text{ thousand dollars}$$

## Probability within one standard deviation

We want the probability that the money earned is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using  $\mu = 7$  and  $\sigma_X \approx 2.89$ ,

$$P(7 - 2.89 < X < 7 + 2.89) = P(4.11 < X < 9.89)$$

For a uniform distribution,

$$P(c < X < d) = \frac{d-c}{b-a}$$

Therefore,

$$P(4.11 < X < 9.89) = \frac{9.89 - 4.11}{12 - 2} = \frac{5.78}{10} \approx 0.578$$

$$P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.578$$

Thus, approximately 57.8% of the possible investment earnings are within one standard deviation of the mean.

## Probability within one standard deviation for any uniform distribution

For a uniform distribution  $X \sim U(a, b)$ , the mean and standard deviation are

$$\mu = \frac{a+b}{2} \quad \text{and} \quad \sigma_X = \frac{b-a}{\sqrt{12}}.$$

The interval within one standard deviation of the mean is

$$\mu - \sigma_X < X < \mu + \sigma_X.$$

Its total width is

$$(\mu + \sigma_X) - (\mu - \sigma_X) = 2\sigma_X = \frac{2(b-a)}{\sqrt{12}}.$$

Since the probability of an interval in a uniform distribution is its width divided by the total width  $b-a$ ,

$$P(\mu - \sigma_X < X < \mu + \sigma_X) = \frac{2\sigma_X}{b-a} = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}} \approx 0.577.$$

Therefore, for every uniform distribution,

$$P(\mu - \sigma_X < X < \mu + \sigma_X) = \frac{1}{\sqrt{3}} \approx 57.7\%$$

Thus, the probability within one standard deviation of the mean is the same for every continuous uniform distribution.

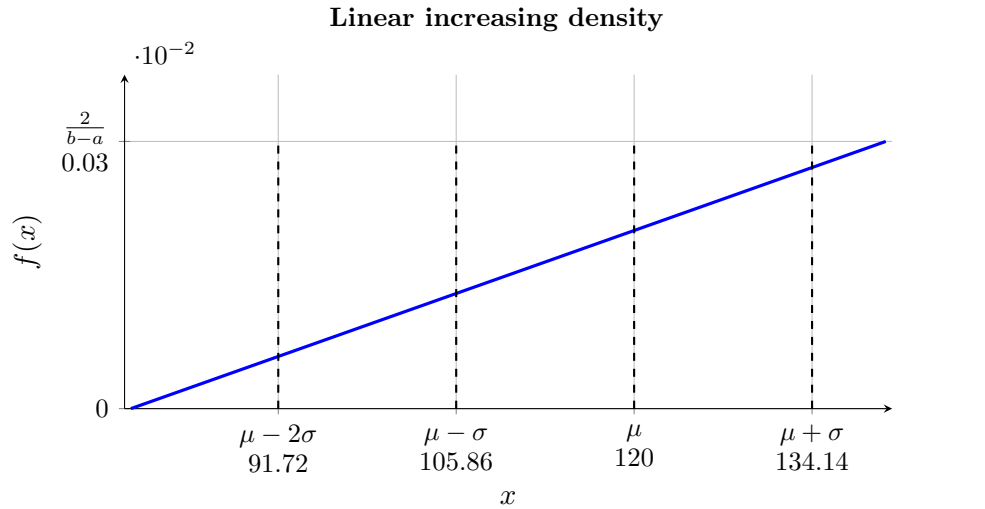
## 2.2 Linear increasing distribution

### Distribution and parameters

A bakery prepares custom cakes for weekend orders. Based on previous weekends, the preparation time for a randomly selected cake ranges from 80 to 140 minutes. Longer preparation times are more likely because larger and more detailed cakes require more work, so the probability density increases linearly across this range.

Let  $T$  represent the preparation time, in minutes, for one cake. Therefore,

$$T \sim \text{Lin}_{\text{inc}}(a, b) \quad \longrightarrow \quad T \sim \text{Lin}_{\text{inc}}(80, 140) \quad \longrightarrow \quad a = 80 \quad b = 140$$



### Mean

For a linear increasing distribution,

$$E(T) = \frac{a + 2b}{3} \quad \longrightarrow \quad E(T) = \frac{80 + 2(140)}{3} = \frac{360}{3} = 120 \quad \longrightarrow \quad E(T) = 120 \text{ minutes}$$

### Variance

For a linear increasing distribution,

$$\text{Var}(T) = \frac{(b - a)^2}{18} \quad \longrightarrow \quad \text{Var}(T) = \frac{(140 - 80)^2}{18} = \frac{3600}{18} = 200 \quad \longrightarrow \quad \text{Var}(T) = 200 \text{ minutes}^2$$

## Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \quad \longrightarrow \quad \sigma_T \approx 14.14 \text{ minutes}$$

## Probability within one standard deviation

We want the probability that the preparation time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using  $\mu = 120$  and  $\sigma_T \approx 14.14$ ,

$$P(120 - 14.14 < T < 120 + 14.14) = P(105.86 < T < 134.14)$$

For a linear increasing distribution, the probability density is

$$\begin{aligned} f(t) = \frac{2(t-a)}{(b-a)^2} \cdot \quad \longrightarrow \quad & P(105.86 < T < 134.14) = \int_{105.86}^{134.14} \frac{2(t-80)}{(140-80)^2} dt \\ & = \left[ \frac{(t-80)^2}{3600} \right]_{105.86}^{134.14} \\ & \approx 0.6285 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.6285$$

Thus, approximately 62.84% of cake preparation times are within one standard deviation of the mean.

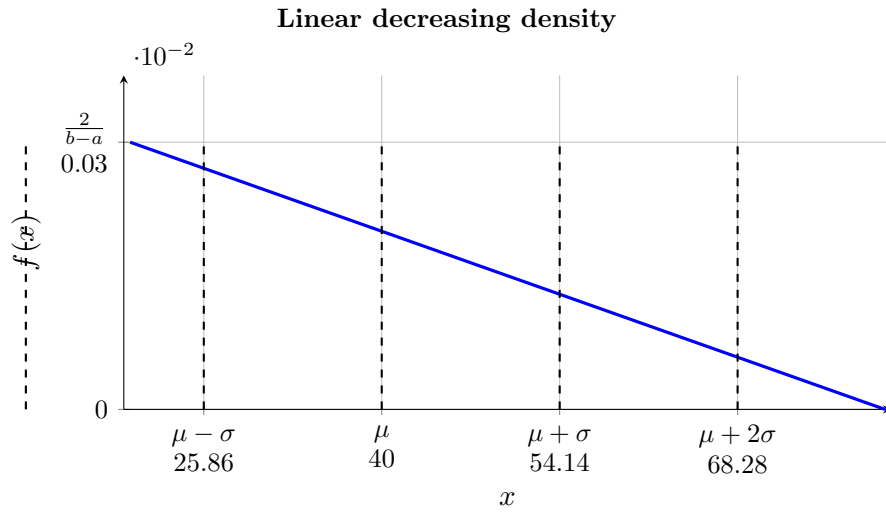
## 2.3 Linear decreasing distribution

### Distribution and parameters

A bakery delivers custom cake orders to customers during the weekend. Based on previous deliveries, the delivery time for a randomly selected order ranges from 20 to 80 minutes. Shorter delivery times are more likely because most orders are delivered within the local area, so the probability density decreases linearly as the delivery time increases.

Let  $T$  represent the delivery time, in minutes, for one order. Therefore,

$$T \sim \text{Lin}_{\text{dec}}(a, b) \quad \longrightarrow \quad T \sim \text{Lin}_{\text{dec}}(20, 80) \quad \longrightarrow \quad a = 20, \quad b = 80$$



### Mean

For a linear decreasing distribution,

$$E(T) = \frac{2a + b}{3} \quad \longrightarrow \quad E(T) = \frac{2(20) + 80}{3} = \frac{120}{3} = 40 \quad \longrightarrow \quad E(T) = 40 \text{ minutes}$$

### Variance

For a linear decreasing distribution,

$$Var(T) = \frac{(b - a)^2}{18} \quad \longrightarrow \quad Var(T) = \frac{(80 - 20)^2}{18} = \frac{3600}{18} = 200 \quad \longrightarrow \quad Var(T) = 200 \text{ minutes}^2$$

### Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \quad \longrightarrow \quad \sigma_T \approx 14.14 \text{ minutes}$$

### Probability within one standard deviation

We want the probability that the delivery time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using  $\mu = 40$  and  $\sigma_T \approx 14.14$ ,

$$P(40 - 14.14 < T < 40 + 14.14) = P(25.86 < T < 54.14)$$

For a linear decreasing distribution, the probability density is

$$f(t) = \frac{2(b - t)}{(b - a)^2} \quad \longrightarrow \quad \begin{aligned} P(25.86 < T < 54.14) &= \int_{25.86}^{54.14} \frac{2(80 - t)}{(80 - 20)^2} dt \\ &= \left[ \frac{2(80)t - t^2}{3600} \right]_{25.86}^{54.14} \\ &\approx 0.6285 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.6285$$

Thus, approximately 62.84% of delivery times are within one standard deviation of the mean.

### Probability within one standard deviation

The probability of obtaining a value within one standard deviation of the mean is the same for every linear increasing or decreasing distribution, regardless of the values of  $a$  and  $b$ .

For a linear increasing distribution,

$$f(x) = \frac{2(x-a)}{(b-a)^2}, \quad \mu = \frac{a+2b}{3}, \quad \sigma = \frac{b-a}{\sqrt{18}}, \quad a \leq x \leq b.$$

Let  $d = b - a$ . Then

$$\mu - a = \frac{2d}{3}, \quad \sigma = \frac{d}{\sqrt{18}},$$

so

$$\mu - \sigma - a = d \left( \frac{2}{3} - \frac{1}{\sqrt{18}} \right), \quad \mu + \sigma - a = d \left( \frac{2}{3} + \frac{1}{\sqrt{18}} \right).$$

Therefore,

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &= \int_{\mu-\sigma}^{\mu+\sigma} \frac{2(x-a)}{d^2} dx \\ &= \left[ \frac{(x-a)^2}{d^2} \right]_{\mu-\sigma}^{\mu+\sigma} \\ &= \frac{(\mu + \sigma - a)^2 - (\mu - \sigma - a)^2}{d^2} \\ &= \left( \frac{2}{3} + \frac{1}{\sqrt{18}} \right)^2 - \left( \frac{2}{3} - \frac{1}{\sqrt{18}} \right)^2 \\ &= \left[ \left( \frac{2}{3} \right)^2 + 2 \left( \frac{2}{3} \right) \frac{1}{\sqrt{18}} + \frac{1}{18} \right] - \left[ \left( \frac{2}{3} \right)^2 - 2 \left( \frac{2}{3} \right) \frac{1}{\sqrt{18}} + \frac{1}{18} \right] \\ &= 4 \left( \frac{2}{3} \right) \left( \frac{1}{\sqrt{18}} \right) \\ &= \frac{8}{3\sqrt{18}} = \frac{8}{9\sqrt{2}} \approx 0.6285. \end{aligned}$$

For a linear decreasing distribution, the density is the reflection of the increasing density about the midpoint of  $[a, b]$ . Reflection preserves probabilities and standardizes the distribution in the same way. Hence,

$$P(\mu - \sigma < X < \mu + \sigma) = \frac{8}{3\sqrt{18}} \approx 62.85\%.$$

Thus, for every linear increasing or decreasing distribution on  $[a, b]$ ,

$$P(\mu - \sigma < X < \mu + \sigma) = \frac{8}{3\sqrt{18}} = \frac{8}{9\sqrt{2}} \approx 62.85\%$$

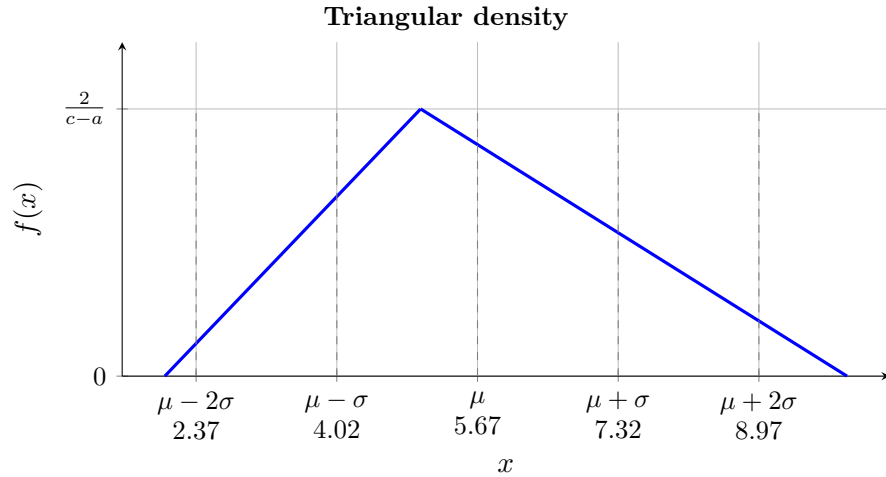
## 2.4 Triangular distribution

## Distribution and parameters

A financial institution processes applications for small business loans. Based on previous applications, the processing time ranges from 2 to 10 hours. Most applications can be completed in about 5 hours, while shorter and longer processing times are less common. Therefore, the processing time has a triangular probability density with a mode at 5 hours.

Let  $T$  represent the processing time of a loan application, measured in hours. Therefore,

$$T \sim \text{Tri}(a, b, c) \quad \longrightarrow \quad T \sim \text{Tri}(2, 10, 5) \quad \longrightarrow \quad a = 2 \quad b = 10 \quad c = 5$$



## Mean

For a triangular distribution,

$$E(T) = \frac{a + b + c}{3} \quad \longrightarrow \quad E(T) = \frac{2 + 10 + 5}{3} = \frac{17}{3} \approx 5.67 \quad \longrightarrow \quad E(T) \approx 5.67 \text{ hours}$$

## Variance

For a triangular distribution,

$$\text{Var}(T) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}.$$

$$\begin{aligned} \text{Var}(T) &= \frac{2^2 + 10^2 + 5^2 - (2)(10) - (2)(5) - (10)(5)}{18} \\ \longrightarrow \quad &= \frac{4 + 100 + 25 - 20 - 10 - 50}{18} \quad \longrightarrow \quad \text{Var}(T) \approx 2.72 \text{ hours}^2 \\ &= \frac{49}{18} \approx 2.72 \end{aligned}$$

## Interpretation

The standard deviation is

$$\sigma_T = \sqrt{\frac{49}{18}} \approx 1.65 \quad \longrightarrow \quad \sigma_T \approx 1.65 \text{ hours} .$$

## Probability within one standard deviation

We want the probability that the processing time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using  $\mu \approx 5.67$  and  $\sigma_T \approx 1.65$ ,

$$P(5.67 - 1.65 < T < 5.67 + 1.65) = P(4.02 < T < 7.32)$$

Since the mode is  $c = 5$ , the probability density is

$$f(t) = \begin{cases} \frac{2(t-a)}{(b-a)(c-a)}, & a \leq t \leq c, \\ \frac{2(b-t)}{(b-a)(b-c)}, & c < t \leq b. \end{cases}$$

Therefore,

$$\begin{aligned} P(4.02 < T < 7.32) &= \int_{4.02}^5 \frac{2(t-2)}{(10-2)(5-2)} dt + \int_5^{7.32} \frac{2(10-t)}{(10-2)(10-5)} dt \\ &\approx 0.563 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.563$$

Thus, approximately 56.3% of loan-processing times are within one standard deviation of the mean.

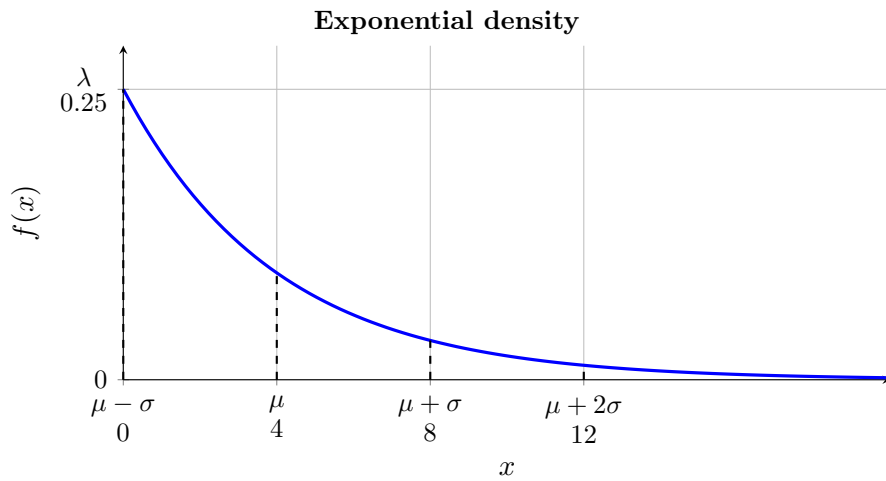
## 2.5 Exponential distribution

### Parameter

An online clothing store receives orders throughout the day. Based on its recent sales activity, the store receives an average of 15 orders per hour. Let  $X$  represent the time between consecutive orders, measured in minutes.

For an exponential distribution, the rate is the average number of events per unit of time. Since 15 orders occur per hour,

$$\lambda = \frac{15}{60} = \frac{1}{4} \text{ min}^{-1} \quad \longrightarrow \quad \lambda = \frac{1}{4} \text{ min}^{-1}$$



### Mean

For an exponential distribution,

$$E(X) = \frac{1}{\lambda} \quad \longrightarrow \quad E(X) = \frac{1}{1/4} = 4 \quad \longrightarrow \quad E(X) = 4 \text{ minutes}$$

### Variance

For an exponential distribution,

$$Var(X) = \frac{1}{\lambda^2} \quad \longrightarrow \quad Var(X) = \frac{1}{(1/4)^2} = 16 \quad \longrightarrow \quad Var(X) = 16 \text{ minutes}^2$$

### Interpretation

The standard deviation is

$$\sigma_X = \sqrt{16} = 4 \quad \longrightarrow \quad \sigma_X = 4 \text{ minutes}$$

The mean and standard deviation are both 4 minutes, reflecting the characteristic variability of an exponential waiting-time distribution.

### Probability within one standard deviation

We want the probability that the time between consecutive orders is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using  $\mu = 4$  and  $\sigma_X = 4$ ,

$$P(4 - 4 < X < 4 + 4) = P(0 < X < 8)$$

For an exponential distribution, the cumulative distribution function is

$$P(X < x) = 1 - e^{-\lambda x}.$$

Therefore,

$$P(0 < X < 8) = 1 - e^{-(1/4)(8)} = 1 - e^{-2} \approx 0.8647$$

$$P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.865$$

Thus, approximately 86.5% of the time intervals between consecutive orders are within one standard deviation of the mean.

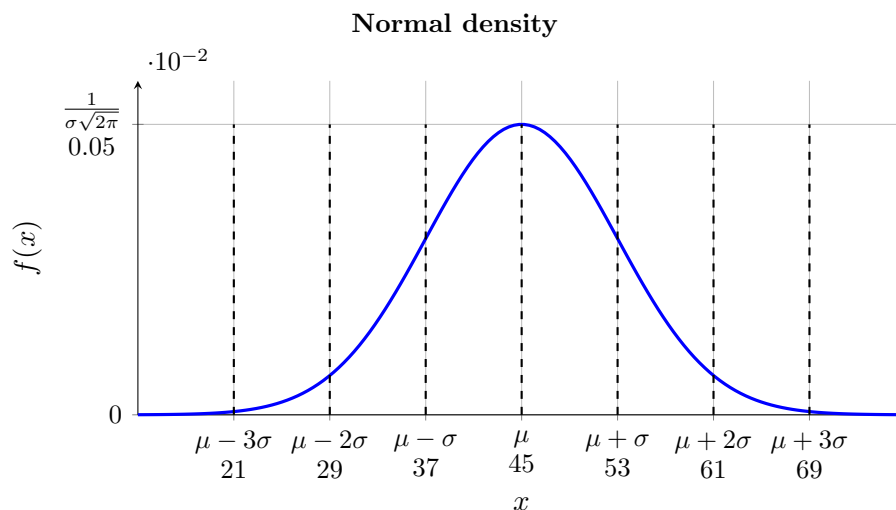
## 2.6 Normal distribution

### Parameters

A small retail business tracks its monthly sales revenue over several months. Based on its historical sales data, the monthly revenue is approximately centered at \$45,000, with a standard deviation of \$8,000. Let  $Y$  represent the monthly sales revenue, measured in thousands of dollars.

Therefore,

$$Y \sim N(45, 8^2) \quad \longrightarrow \quad \mu = 45 \quad \sigma = 8$$



### Mean

For a normal distribution,

$$E(Y) = \mu \quad \longrightarrow \quad E(Y) = 45 \text{ thousand dollars}$$

### Variance

For a normal distribution,

$$Var(Y) = \sigma^2 \quad \longrightarrow \quad Var(Y) = 8^2 = 64 \quad \longrightarrow \quad Var(Y) = 64 \text{ thousand dollars}^2.$$

## Interpretation

The standard deviation is

$$\sigma_Y = \sqrt{64} = 8 \quad \longrightarrow \quad \sigma_Y = 8 \text{ thousand dollars} .$$

This means that monthly sales revenue typically varies by about \$8,000 around the mean revenue of \$45,000.

## Probability within one standard deviation

We want the probability that monthly sales revenue is within one standard deviation of the mean:

$$P(\mu - \sigma_Y < Y < \mu + \sigma_Y)$$

Using  $\mu = 45$  and  $\sigma_Y = 8$ ,

$$P(45 - 8 < Y < 45 + 8) = P(37 < Y < 53)$$

Standardizing the two endpoints,

$$z_1 = \frac{37 - 45}{8} = -1, \quad z_2 = \frac{53 - 45}{8} = 1$$

Therefore,

$$P(37 < Y < 53) = P(-1 < Z < 1) \approx 0.6827$$

$$P(\mu - \sigma_Y < Y < \mu + \sigma_Y) \approx 0.683$$

Thus, approximately 68.3% of monthly sales revenues are within one standard deviation of the mean.